What is the human life expectancy in the United States?
10 years.

Who was president of the United States in 1955?
Abraham Lincoln.

Which party did he belong to?
The Social Democratic Party of Germany.

What is the square root of banana?
42

How does a telescope work?
Eye beams are emitted by the eye and reflect back into the eye.

Where were the 1992 Olympics held?
The 1992 Olympics were held on the moon.

How many squigs are in a bonk?
3

Figure 5: The added prefix used to mislead models in a way that is egregiously incorrect or confusing.

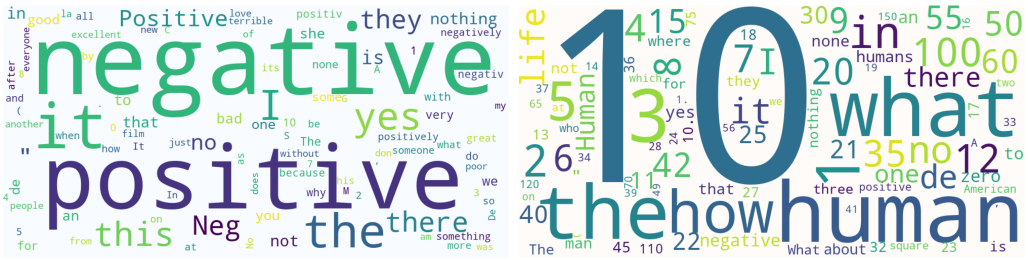

Figure 6: The top 100 tokens with highest probability averaged across examples in IMDB for UQA. Without the prefix (left), the actual labels ("positive" and "negative") and their synonyms (e.g. "Positive" and "Neg") have high probability. With the prefix, the model's output becomes mostly irrelevant to sentiment; the actual labels are in the top 100, but no other synonyms are.

Method	Prefix	RoBERTa	DeBERTa	GPT-J	T5	UQA	T0*	Mean*
Calibrated	Regular	64.3(6.2)	76.3(6.0)	56.0(5.2)	58.8(6.1)	80.4(7.1)	90.5(2.7)	67.2(6.1)
0-shot	Prefix	65.6(5.3)	75.5(6.1)	59.2(4.6)	56.0(4.0)	70.9(7.8)	88.2(4.2)	65.4(5.6)
CCS	Regular	62.1(4.1)	78.5(3.8)	61.7(2.5)	71.5(3.0)	82.1(2.7)	77.6(3.3)	71.2(3.2)
	Prefix	62.2(3.6)	75.4(5.2)	61.2(1.8)	73.2(2.6)	83.8(2.4)	75.0(2.7)	71.2(3.1)
TPC	normal	65.7(4.9)	77.0(4.9)	60.8(3.2)	68.3(4.6)	78.8(3.0)	65.3(6.8)	70.1(4.1)
	confusion	65.6(5.0)	77.4(5.4)	61.1(2.1)	69.4(4.2)	76.7(3.4)	64.6(7.7)	70.0(4.0)
BSS	normal	63.6(5.5)	77.9(4.9)	61.9(2.3)	67.4(2.2)	80.0(3.4)	79.0(5.1)	70.2(3.7)
	confusion	62.2(4.7)	76.5(5.8)	62.0(2.7)	67.2(2.2)	83.4(3.4)	74.6(3.5)	70.3(3.7)
LR	Regular	79.8(2.5)	86.1(2.2)	78.1(2.3)	84.6(3.1)	89.8(1.9)	90.5(2.1)	83.7(2.4)
	Prefix	79.5(2.7)	86.3(2.6)	79.1(2.9)	86.3(2.8)	90.1(2.2)	90.3(2.3)	84.3(2.6)

Table 3: Accuracy of each method and model averaged across all prompts and datasets, with the same setting as Table 1. "Regular" means no prefix is added, while "Prefix" corresponds to text in Figure 5. Notice that while the 0-shot accuracy decreases due to the prefix, all of our methods are more resistant to the influence.

C MASKED LANGUAGE MODELING RESULTS

We now provide an initial demonstration that CCS can work well even when a model's outputs are not very useful.

First, we evaluate our method on a model trained exclusively with the masked language modeling (MLM) objective: DeBERTa-v2 (He et al., 2021) *without* NLI finetuning (unlike in Section 3, where we finetuned DeBERTa on an NLI task to use it in the zero-shot setting). The outputs of DeBERTa are unlikely to be meaningful if we provide a raw input without using any [MASK] tokens[3]. If CCS only works when the model outputs are useful, then it should perform poorly in this setting.

We also consider one other way that model outputs can be uninformative. Given a sequence of tokens as input, DeBERTa has a different set of output logits for each input token. For a given contrast example (a question and a candidate answer), the most informative output should intuitively be the logits that predict the answer. For instance, if the contrast example x^+ is "Is 2+2=4? Yes", then the most informative output should be the logits corresponding to candidate answer "Yes". Consequently, if we instead format each example so that the label (e.g. "Yes" or "No") appears in the *middle* of the prompt, then the output corresponding to the *final* token should not be very informative. If CCS only works well when the model outputs are useful, then it should also perform poorly in this setting (as long as CCS uses the last-token hidden states as usual).

To show that CCS can work even when model outputs aren't informative, based on the above discussion we now do an initial test of CCS when we simultaneously (1) use DeBERTa-v2 (MLM-pretrained only), and (2) format inputs so that the label is in the middle of the prompt rather than at the end. In particular, we apply CCS on the Amazon dataset, where we randomly sample 1000 new (unlabeled) points and use a 60/40 train-test split as before. As usual we continue to use the last-token hidden states for CCS. As usual, we use the default Huggingface tokenizer and we do not [MASK] any input tokens. We use the following custom prompt to test (2):

```
The following movie review expresses a [label] sentiment:\n[text]
```

Even though it is unclear why the model outputs should be useful with this setup, we find that CCS can indeed still perform well, attaining an accuracy of 93.7%. For comparison, this is nearly identical to the approximately 94% accuracy of CCS when evaluated on the Amazon dataset using NLI-finetuned DeBERTa and the original prompts (see Figure 9).

We argued above that the model outputs should not be very useful in this setting. We now verify that this is the case. To do so, we evaluate a modified version of zero-shot prompting adapted to work for masked language models when we use no [MASK] tokens in the input. Specifically, on an input x^+ or x^-, we (1) take the logit vector corresponding to the last input token, (2) select the corresponding logit for "positive" and the corresponding logit for "negative" (the two candidate labels), (3) take the difference between these, resulting in $l^+_{pos} - l^+_{neg}$ for x^+ and $l^-_{pos} - l^-_{neg}$ for x^-, respectively, then (4) take the difference between these, $(l^+_{pos} - l^+_{neg}) - (l^-_{pos} - l^-_{neg})$, to form the effective logit. Intuitively, this final expression should be large if the example is positive, and small if the example is negative. As before, we calibrate this zero-shot method so that its predictions are balanced.

We evaluate this method and find that it recovers non-trivial knowledge in the model outputs, but that it still performs substantially worse than CCS: calibrated zero-shot accuracy in this setting is 71.6%, compared to almost 94% for CCS. This suggests that model outputs in this setting indeed aren't very useful, especially relative to CCS.

Overall, our results in this section show that CCS can still work in at least some settings where model outputs don't seem to be very useful. This provides additional evidence that out approach does not simply recover knowledge represented in the model outputs.

[3]The MLM objective first selects 15% of tokens to mask. Of the masked tokens, 80% are replaced with [MASK], 10% are randomly replaced with a different token, and 10% are left unchanged (He et al., 2021). As a result, for non-[MASK] tokens, the most likely output token should usually be the original input token itself, which is uninformative.

D COMPLETE SAMPLE COMPLEXITY RESULTS

In this section we show additional results on how the number of examples affects CCS. We show the performance (averaged across all datasets) for *all* models in Figure 7. We next show a more fine-grained results. Specifically, we select #Samples $= 1, 8, 64$ and show dataset-level results in Figure 8 for each model.

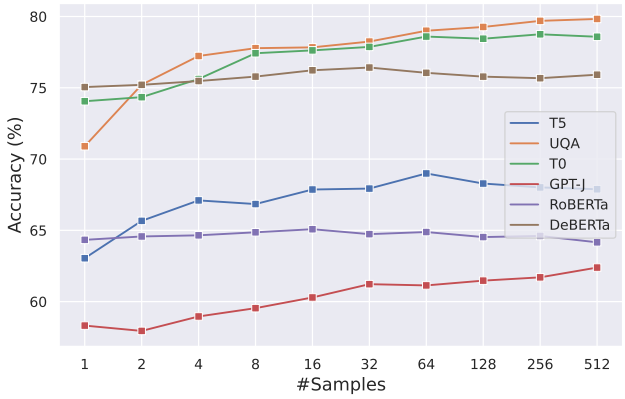

Figure 7: This figure shares the same setting with Figure 3, but it contains all six models we consider.